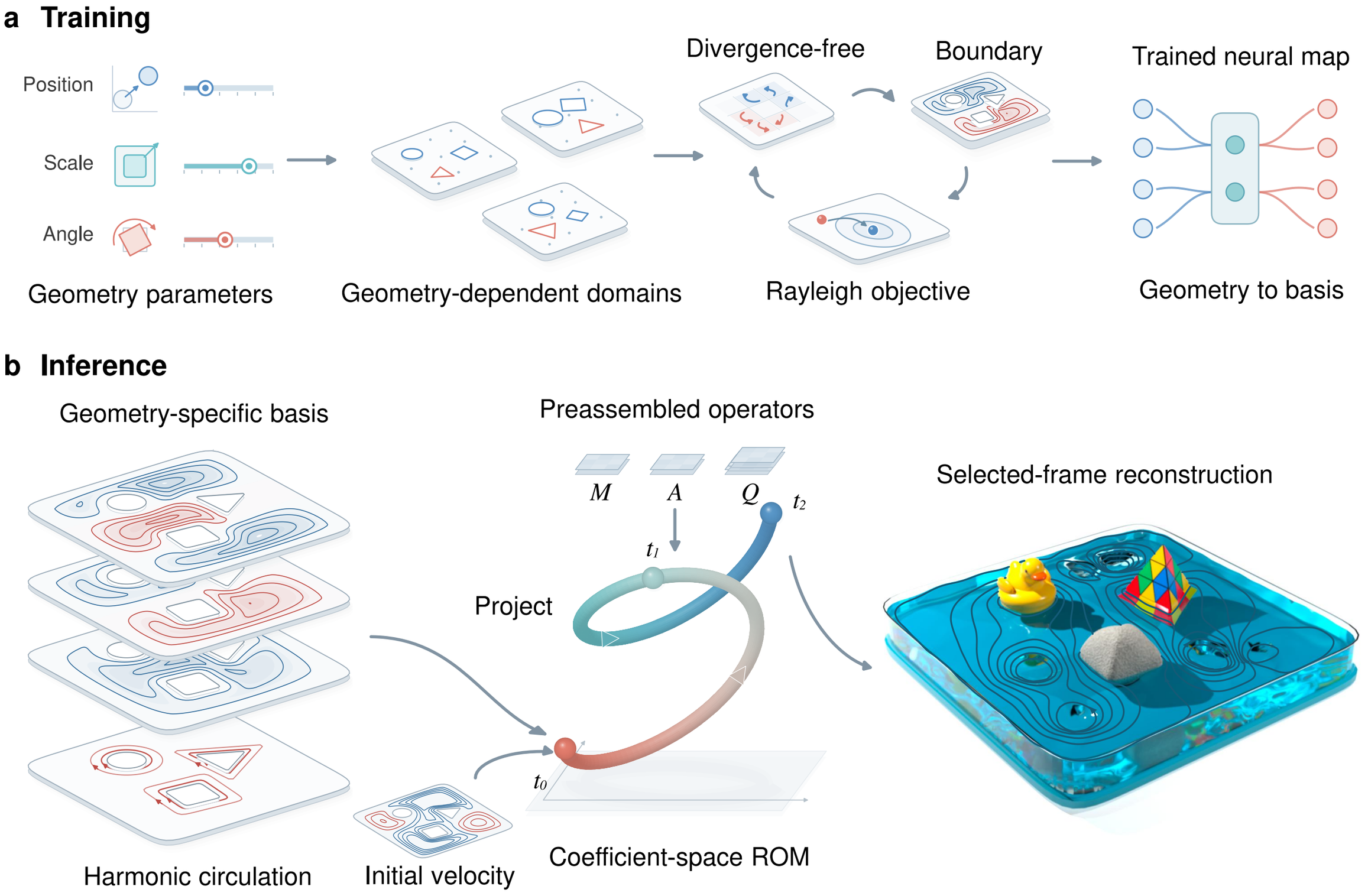

a Training
Position
Scale
Angle
Geometry parameters
Geometry-dependent domains
Divergence-free
Boundary
Rayleigh objective
Trained neural map
Geometry to basis
b Inference
Geometry-specific basis
Harmonic circulation
Initial velocity
Preassembled operators
M
A
Q
t2
t1
Project
t0
Coefficient-space ROM
Selected-frame reconstruction